Kalyan Kumar Radhakrishnan Santhakumari

Technologist | INAF - Osservatorio Astronomico di Padova, Italy

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PROFILE

Research and engineering work in adaptive optics and ground-based astronomical instrumentation, with a focus on system engineering, assembly, integration and verification (AIV), optical alignment and telescope commissioning.

APPOINTMENTS

2026 - present	**Technologist, III level** INAF - Osservatorio Astronomico di Padova, Italy (permanent appointment).
2019 - 2026	**Researcher, III level** INAF - Osservatorio Astronomico di Padova, Italy.
2017 - 2018	**Postdoctoral researcher** Max Planck Institute for Astronomy, Heidelberg, Germany.

EDUCATION

2013 - 2017	**PhD, Astronomy and Astrophysics** Heidelberg University / Max Planck Institute for Astronomy, Germany. Research on LINC-NIRVANA adaptive optics.
2011 - 2013	**Erasmus Mundus MSc, Astronomy and Astrophysics** AstroMundus programme (Innsbruck, Padova and Göttingen).
2006 - 2011	**Integrated MSc, Physics** Pondicherry University, India.

SELECTED INSTRUMENT WORK

SOXS | New Technology Telescope, La Silla | 2020 - present
AIV Lead and Manager; Deputy System Engineer. Integration and verification in Padova; installation, acceptance and commissioning activities at the NTT.

MAVIS | Very Large Telescope | 2019 - present
Work-package responsibilities for adaptive-optics calibration strategy, diagnostics and non-common-path aberration calibration; support to design reviews.

MORFEO | Extremely Large Telescope | 2024 - present
Contributions to AIV planning, including the planning document for integration and testing at the ELT Nasmyth platform.

Selected instrument work (continued)

NIRVANA-VIS | Large Binocular Telescope | 2022 - present
System engineering and AIV planning for the visible-wavelength upgrade; PhD co-supervision.

INGOT wavefront sensor | 2019 - present
Laboratory integration, alignment and testing of a wavefront sensor for laser guide stars; student supervision.

LINC-NIRVANA | Large Binocular Telescope | 2013 - 2023
Wavefront-sensor integration, adaptive-optics development, telescope commissioning and early-science support.

SHARK-NIR | Large Binocular Telescope | 2019 - 2024
Instrument AIV support, deformable-mirror characterization and commissioning-plan development.

TEACHING AND SUPERVISION

Teaching in adaptive optics and optical/infrared instrumentation at the University of Padova; earlier wavefront-sensing laboratory teaching at Heidelberg University. Co-supervision of doctoral research in instrumentation and wavefront sensing.

TECHNICAL AREAS

Adaptive optics and wavefront sensing; optical metrology and alignment; instrument integration and verification; calibration; astronomical detectors; commissioning and operational procedures. Programming and analysis: Python, IDL, C and C++.

SELECTED PUBLICATIONS

2024 **Ragazzoni, R., et al.** Ingot-like class of wavefront sensors for laser guide stars. Astronomy & Astrophysics, 688, A21.

2024 **Santhakumari, K. K. R., Battaini, F., et al.** What is your favorite transient event? SOXS is almost ready to observe! Proceedings of SPIE, 13096, 1309673.

2022 **Santhakumari, K. K. R., Battaini, F., et al.** From Assembly to the Complete Integration and Verification of the SOXS Common Path. Proceedings of SPIE, 12184, 1218482.

2019 **Santhakumari, K. K. R., et al.** Operation of a layer-oriented multiconjugate adaptive optics system in the partial illumination regime. Journal of Astronomical Telescopes, Instruments, and Systems, 5(4), 049002.

2016 **Kasper, M., Santhakumari, K. K. R., et al.** New circumstellar structure in the T Tauri system. Astronomy & Astrophysics, 593, A50.

A fuller publications record is available through the linked ORCiD and Google Scholar profiles.